

Project Design Phase-II

Solution Requirements (Functional & Non-functional)

Date	11 July 2026
Team ID	SWTID-2026-9508
Project Name	Credit Card Approval Prediction System
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)
FR-1	Applicant Data Input	Web form to enter applicant demographic and financial details Input validation for required fields
FR-2	Credit Approval Prediction	Preprocess submitted data using saved encoder/scaler Generate Approved / Rejected prediction using trained ML model
FR-3	Result Display	Display prediction result to the user on a results page Show basic explanation of key input factors
FR-4	Model Training Pipeline	Load and clean historical credit card approval dataset Train and evaluate multiple ML models (Logistic Regression, Decision Tree, Random Forest, XGBoost)
FR-5	Model Comparison & Selection	Compare trained models using Accuracy, Precision, Recall, F1-Score Persist the best-performing model as model.pkl
FR-6	Error Handling	Handle invalid or missing form input gracefully Display a custom 404 page for invalid routes

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description
NFR-1	Usability	The web interface should be simple enough for a bank officer with no technical background to enter applicant details and read the prediction result without training.
NFR-2	Security	Applicant financial data submitted through the form should not be logged or stored beyond what is required to generate a prediction.
NFR-3	Reliability	The model should return a valid Approved/Rejected prediction for any correctly formatted input, with graceful handling of missing or invalid fields.
NFR-4	Performance	The application should return a prediction within 2-3 seconds of form submission on standard hardware.
NFR-5	Availability	The Flask application should remain accessible during standard testing and demonstration sessions, with minimal downtime.
NFR-6	Scalability	The prediction pipeline (preprocessing + model inference) should be reusable and extendable to handle additional applicant features in the future.